Beyond Economic Growth

How Climate Resilience, Agriculture, and Trade Shape Food Security

An interactive data visualization exploring how economic growth, climate change, agricultural capacity, and trade together influence food security across 177 countries between 2010 and 2023.

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THE QUESTION

Does Wealth Guarantee a Full Plate?

Economic growth is widely viewed as the foundation of food security. This project challenges that assumption by examining how climate stress, agricultural capacity, and trade dependency influence food availability—and whether prosperity alone is enough to build resilient food systems.

ECONOMY

Does higher GDP per capita consistently translate into greater food availability?

CLIMATE

Which countries maintain food security despite rising temperatures?

AGRICULTURE

Do agricultural land and production improve food security?

RESILIENCE

Which countries balance prosperity, climate resilience, and food security most effectively?

STEP 1 — DATASET

Five Datasets. Two Trusted Sources. One Integrated Panel.

177	14	5	2
COUNTRIES	YEARS (2010–2023)	INTEGRATED DATASETS	DATA PROVIDERS

FAOSTAT

- Food Balance Sheets (Daily calorie supply)
- Temperature Change on Land

World Bank

- GDP per Capita (current US$)
- Population, total
- Agricultural Land (% of land area)

Data Preparation Highlights

- Removed global and regional aggregates
- Standardized country names across sources
- Eliminated duplicate country records
- Reconstructed production, imports, and exports from commodity-level data

STEP 2 — ANALYTICAL QUESTIONS

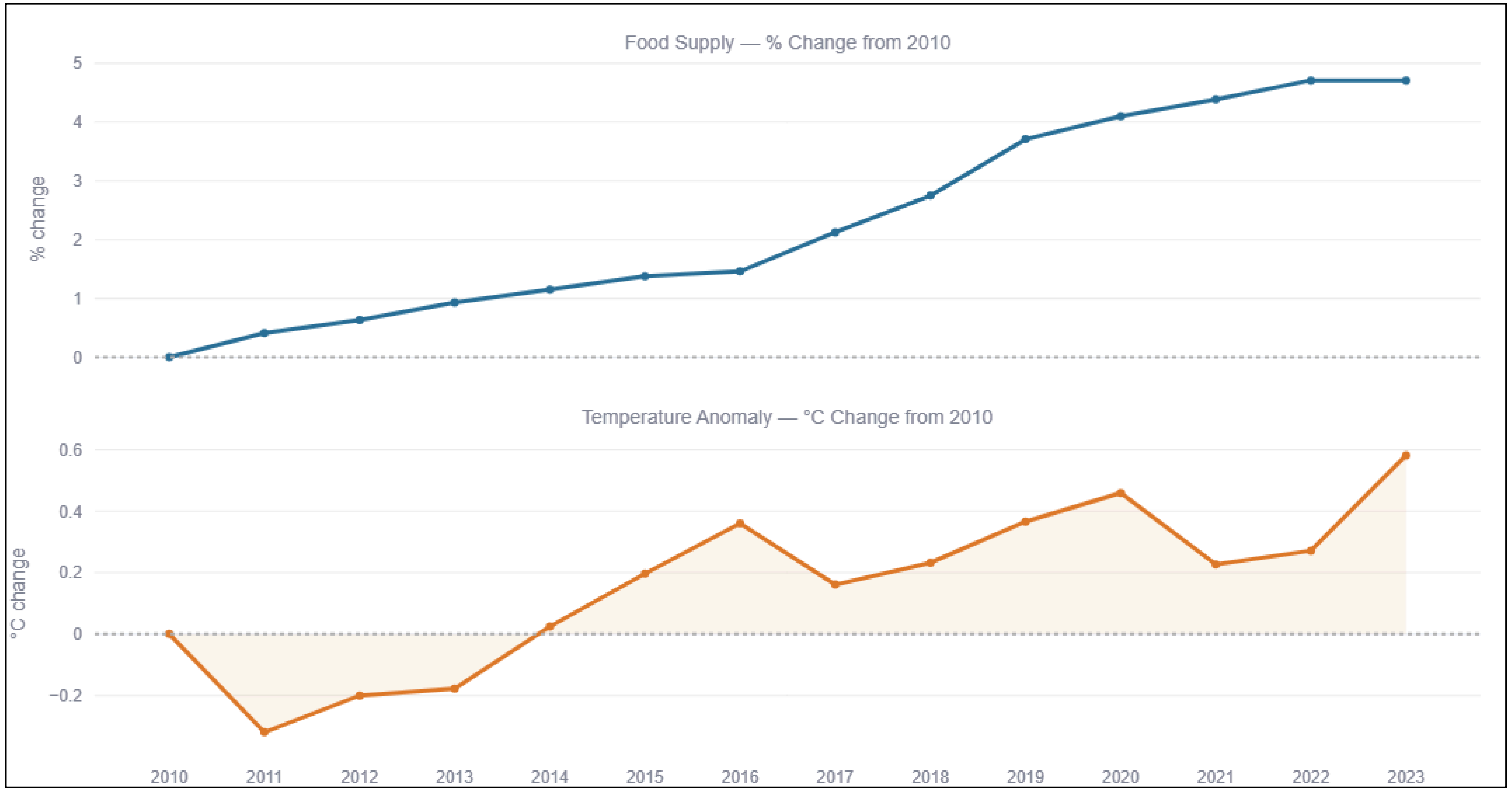
Twelve Questions, Four Themes

The analytical questions investigate relationships, comparisons, trends, and resilience across multiple dimensions of food security.

Economy & Prosperity	Climate & Resilience	Agriculture & Trade	Change & Composite View
AQ1 — Does GDP predict food availability?	AQ2 — How is temperature change associated with food availability?	AQ5 — Which food groups contribute most to global calorie supply?	AQ10 — Which countries improved food security the most since 2010?
AQ6 — How does import dependency influence food security?	AQ3 — Which has a stronger relationship with food availability: GDP or climate?	AQ7 — Does higher food production improve domestic food availability?	AQ11 — Global food supply vs. climate, 2010–2023
AQ9 — Has economic growth translated into food security gains over time?	AQ4 — Which has a stronger relationship with food availability: GDP or climate?	AQ8 — Does more farmland improve food security?	AQ12 — Which countries rank highest on the composite Resilience Index?

FINDING 01 · GLOBAL TRENDS

Climate Stress Is Outrunning Food Supply Gains



+4.7%

GLOBAL CALORIE SUPPLY SINCE 2010

+0.58°C

TEMPERATURE ANOMALY SINCE 2010

Food supply has grown steadily but modestly. Climate stress has accelerated far faster and far less predictably — the two trajectories are not remotely in balance.

FINDING 02 · ECONOMY & FOOD

More Wealth Doesn't Always Mean More Food

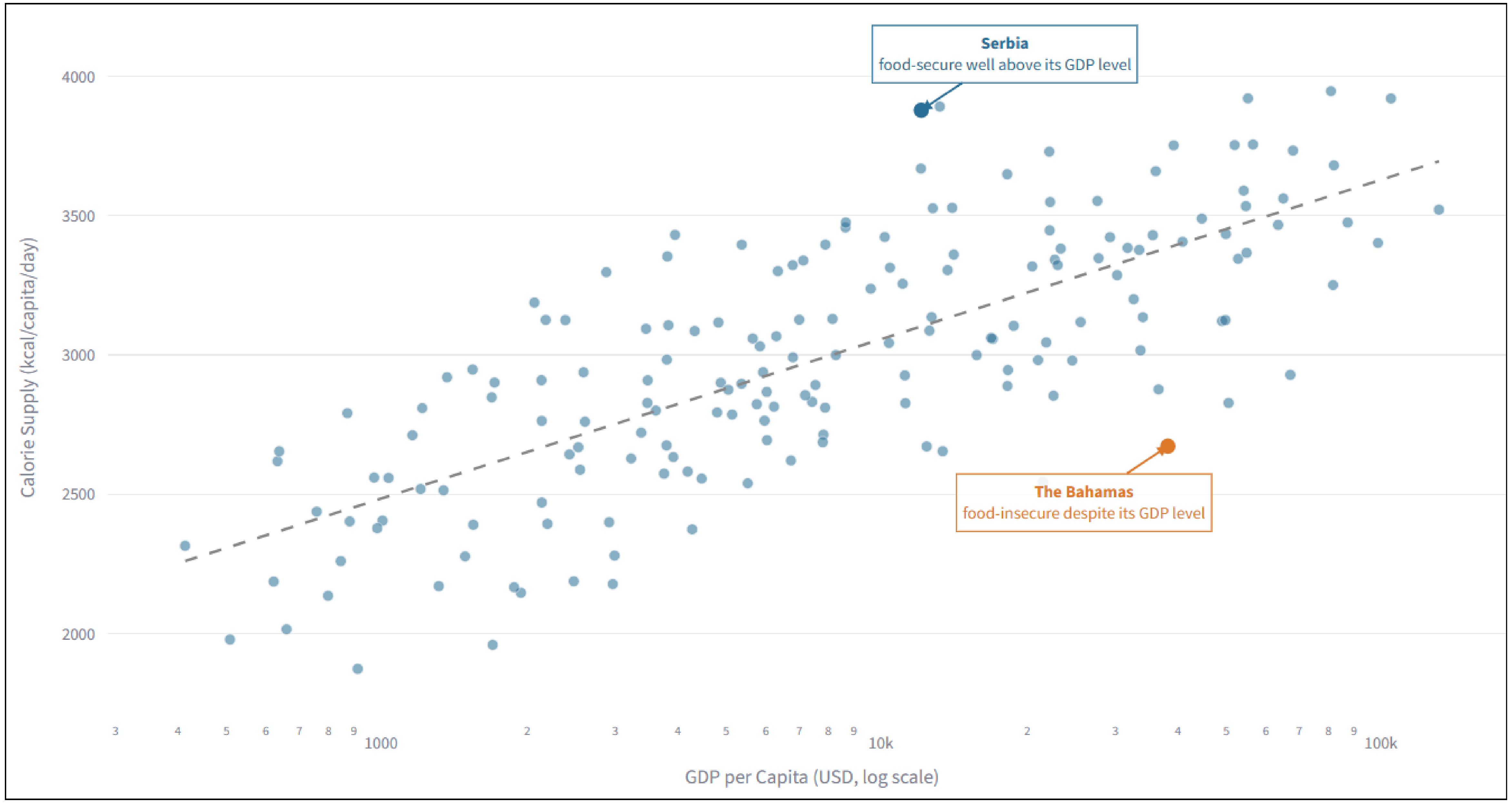
0.75

GDP – FOOD CORRELATION

56%

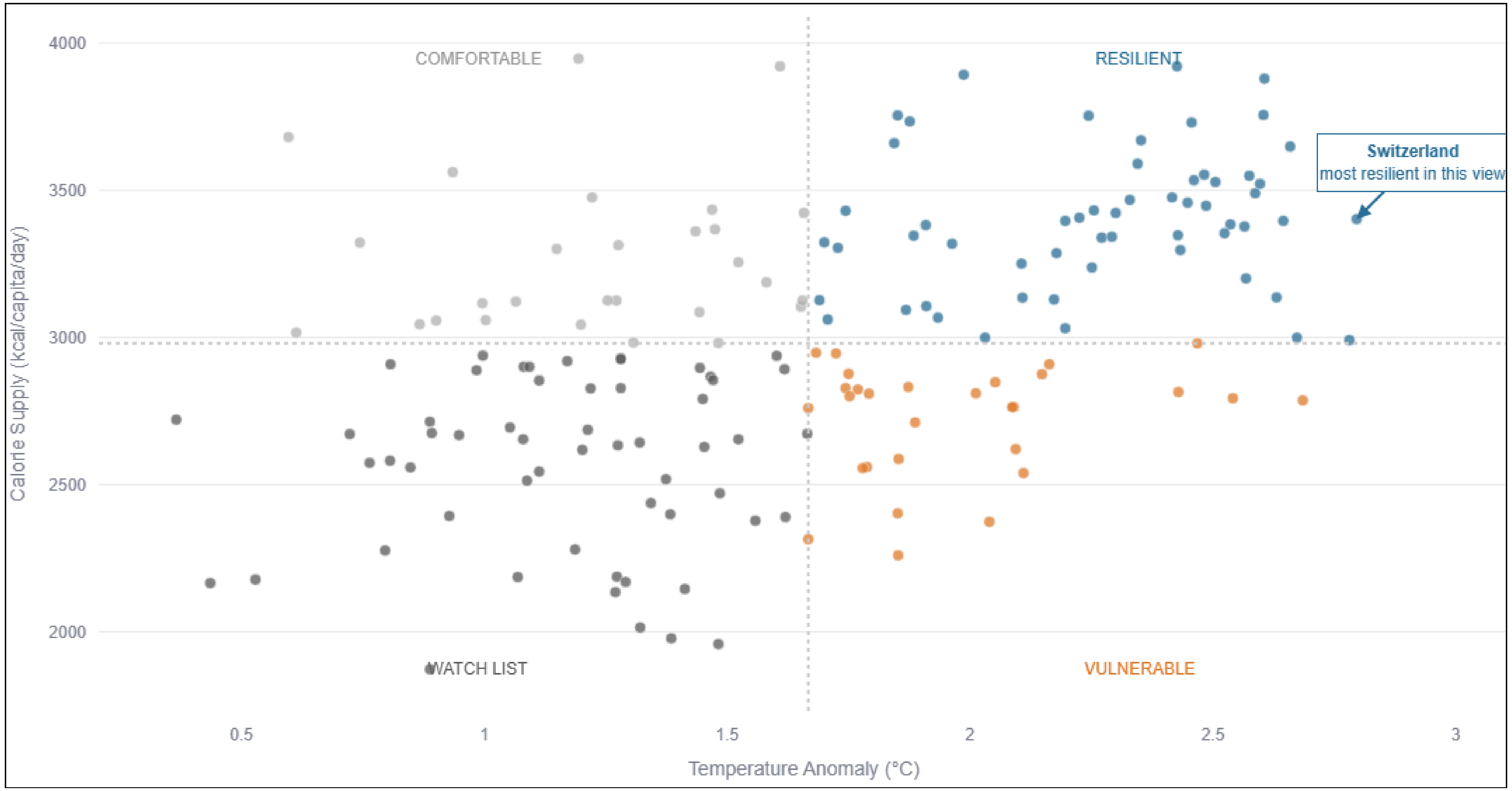
OF FOOD AVAILABILITY VARIATION EXPLAINED BY GDP

Serbia is food-secure well above its GDP level. The Bahamas is food-insecure despite a high GDP. GDP explains a majority of the story — but the other 44% is exactly what climate, agriculture, and trade explain next.



FINDING 03 · CLIMATE IMPACT

Some Countries Beat the Climate Penalty



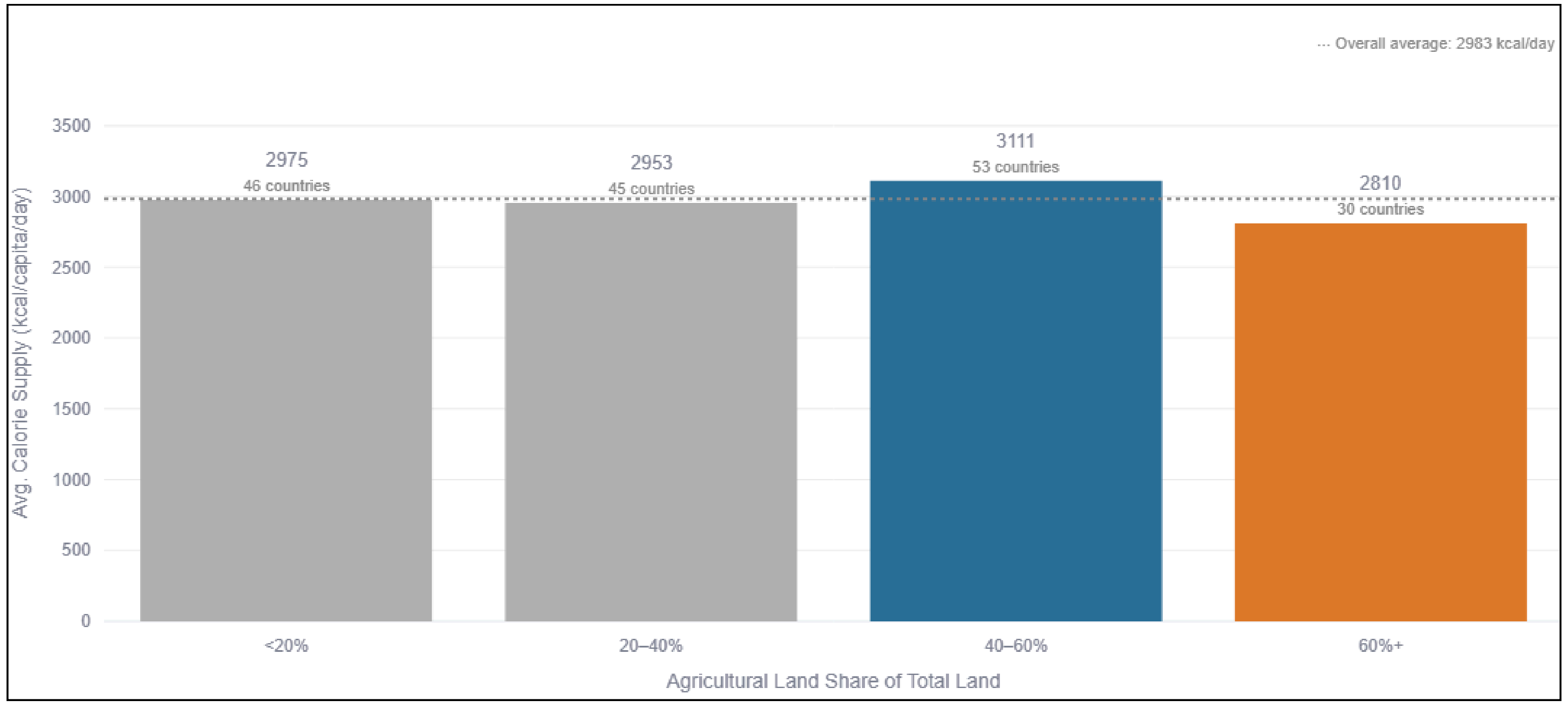
34%

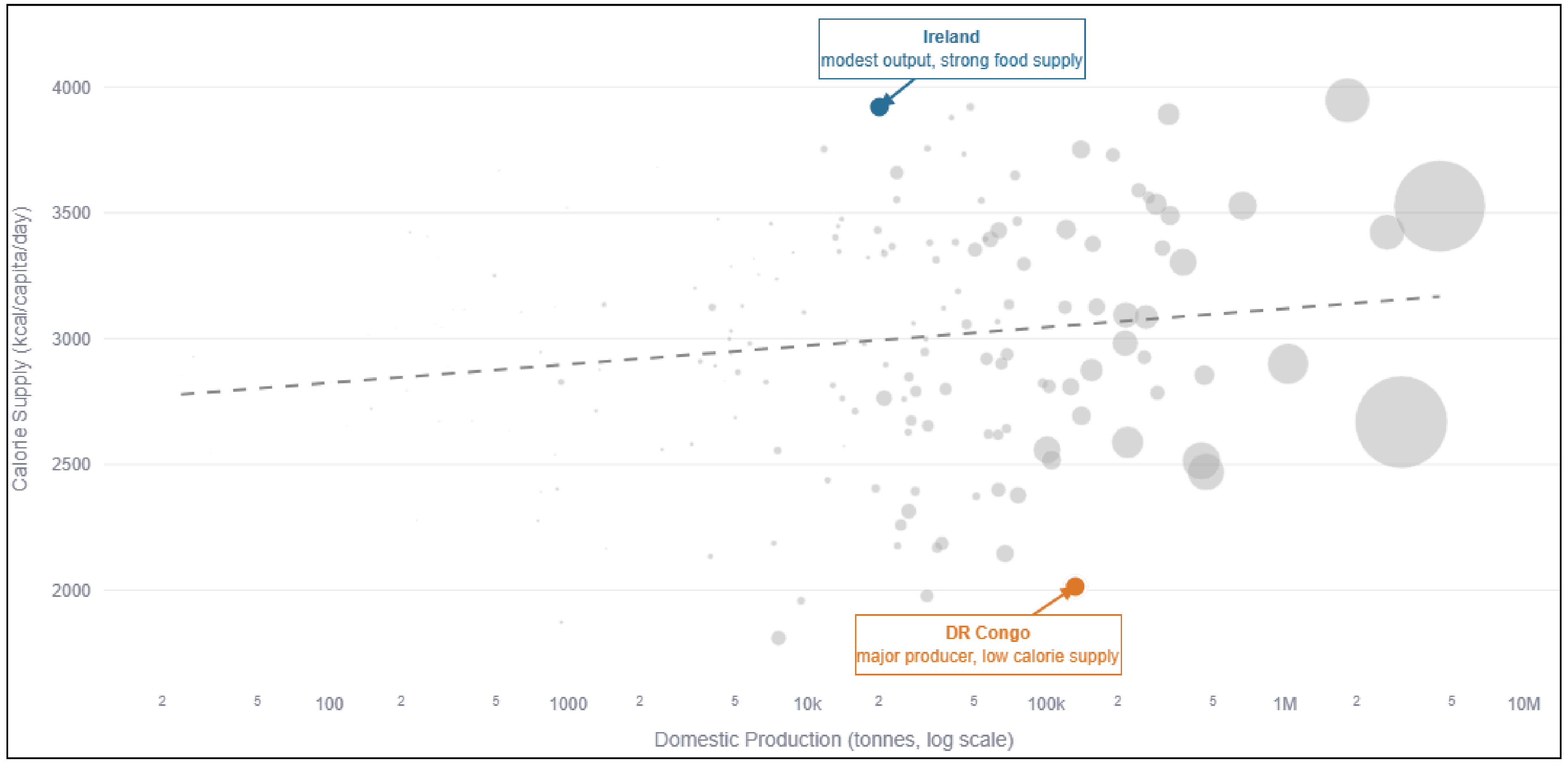
OF COUNTRIES ARE CLIMATE-"RESILIENT"

Countries are split into four groups by warming and food supply. Switzerland stands out as the most resilient — high warming, high food security. Climate stress alone doesn't decide the outcome; what a country does about it — agriculture, trade, infrastructure — matters just as much.

FINDING 04 · AGRICULTURE & PRODUCTION

Land and Output Aren't the Answer Either





More farmland ≠ more food

Land–food correlation: –0.04 — essentially none.

More output ≠ more food at home

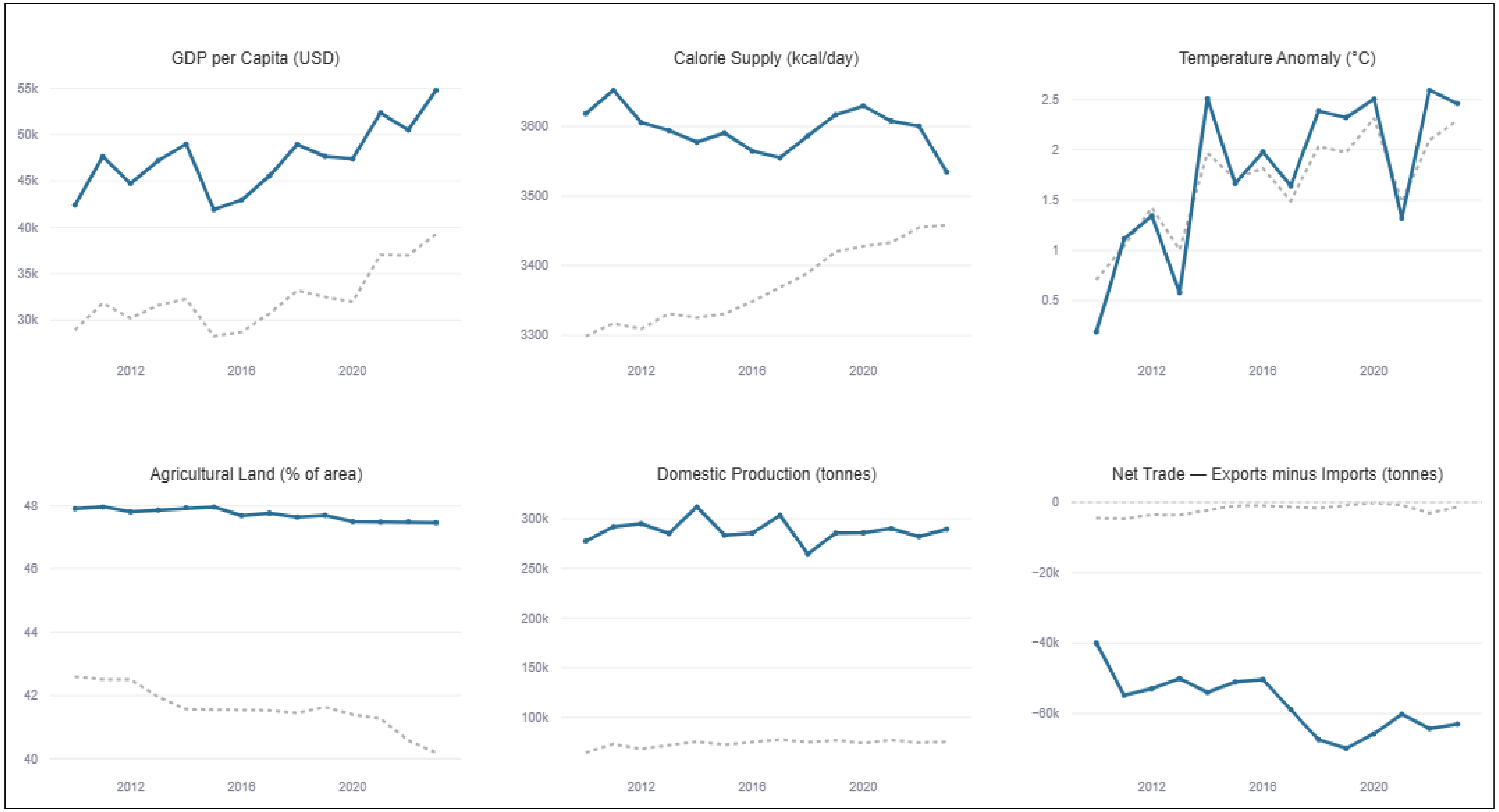
DR Congo: major producer, low calorie supply. Ireland: modest output, strong food supply.

Production feeds trade as much as domestic tables — a large share of what's produced can be exported rather than consumed at home.

FINDING 05 · COUNTRY EXPLORER

One Country, Diminishing Returns

Explore Every Country Through a Single Interactive View - Australia shown as an example



GDP PER CAPITA

+331%

vs. regional average

CALORIE SUPPLY

+20%

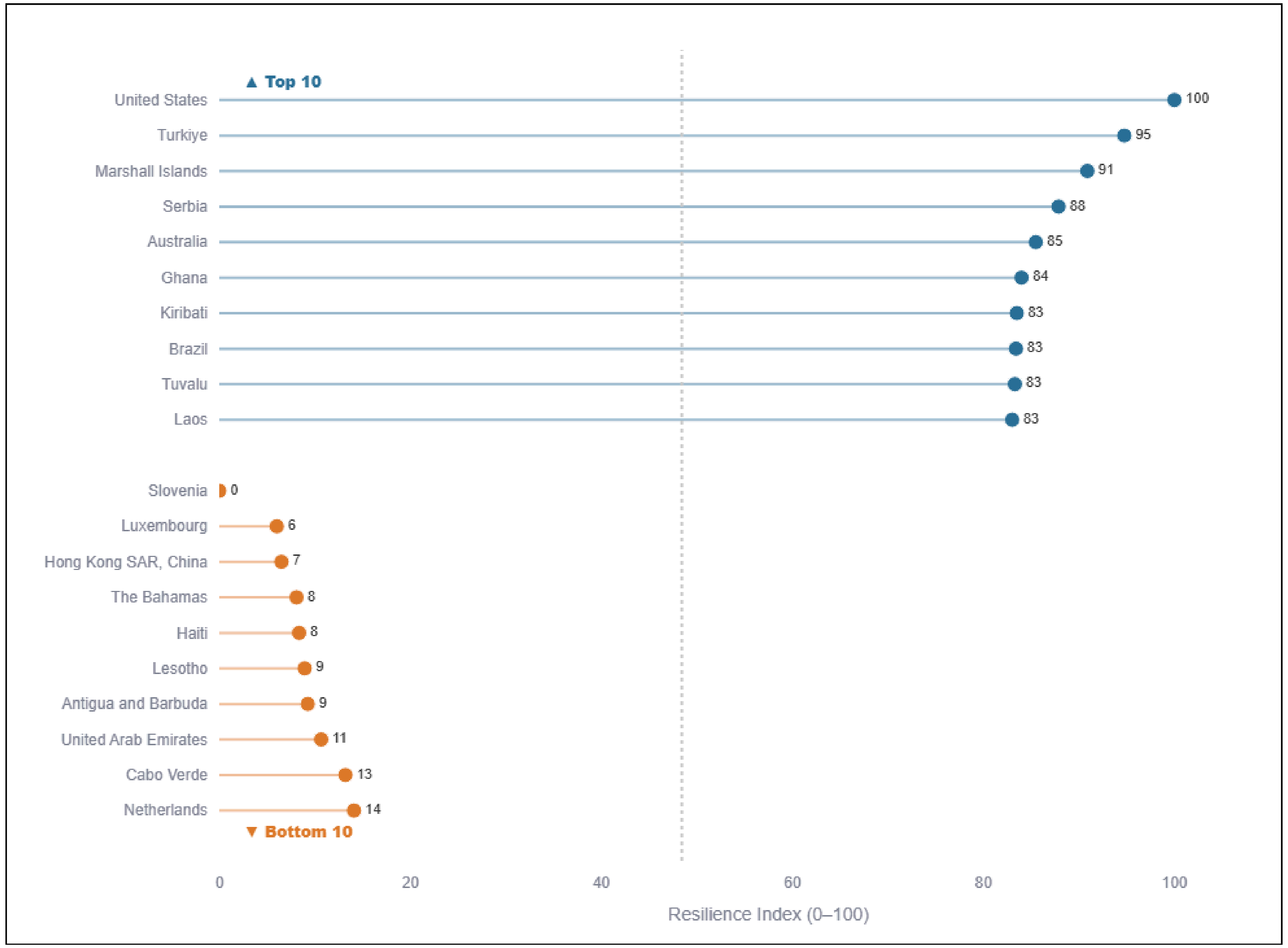
vs. regional average

A 331% increase in GDP per capita corresponded with only a 20% higher calorie supply relative to the regional average.

FINDING 06 · RANKINGS

One Score, Four Factors: the Resilience Index

The Resilience Index combines four factors from this dashboard — food supply, climate stress, GDP-adjusted performance, and trade self-sufficiency — into a single, equally-weighted, outlier-resistant score from 0 to 100.



#1

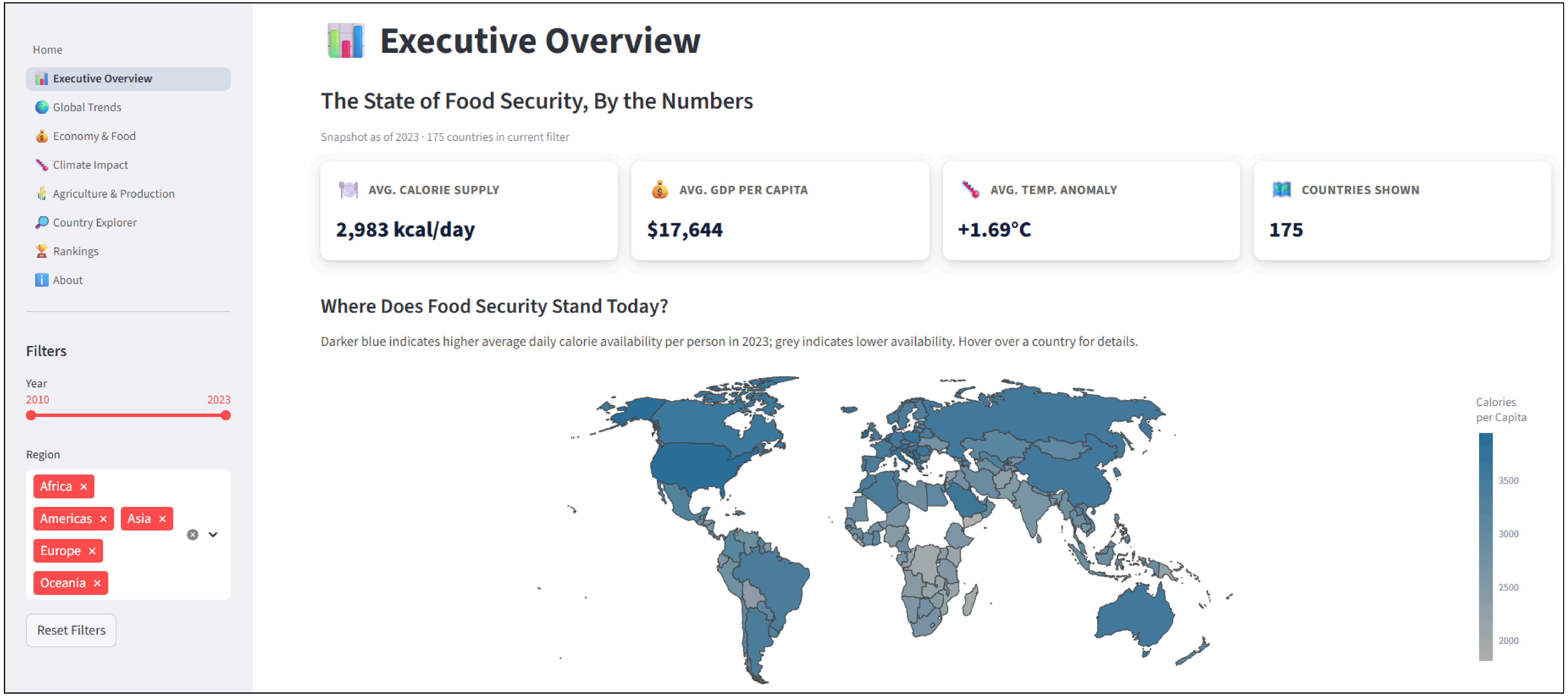
UNITED STATES — TOP RESILIENCE SCORE

Yet it ranks only #7 globally on GDP per capita — resilience isn't purchased by wealth alone.

That's this dashboard's thesis in one number: resilience comes from balancing economic strength, climate exposure, and self-sufficiency — not from wealth alone.

THE DELIVERABLE

A Live, Interactive Dashboard — Not Just Static Slides



7 Dashboard Pages

2+ Global Filters

1 Interactive country-level analysis

DESIGN PRINCIPLES

Built the Way a BI Team Would Ship It

CVD-Safe Palette

Muted blue, amber, and grey improve readability while remaining accessible to users with color vision deficiencies.

Takeaway-First Titles

Each chart title communicates the primary insight rather than simply describing the variables shown.

Direct Annotation

Key countries and outliers are labelled directly on the charts, reducing reliance on legends.

Genuine Interactivity

Interactive filters and the Country Explorer allow users to investigate findings from multiple perspectives.

Consistent Hierarchy

Consistent layouts, typography, KPI cards, and spacing create a coherent dashboard experience.

Transparent Methodology

Composite indices, assumptions, and data transformations are clearly documented rather than hidden.

CONCLUSIONS

The Story Behind Food Security Is More Than Growth

1

GDP explains a majority of food security — but not all of it.

GDP is a strong predictor of food availability—but not a complete explanation.
A correlation of 0.75 and R^2 of 56% indicate that climate, agriculture, and trade contribute substantially to food security outcomes.

2

Climate stress is outpacing food-system adaptation.

Global temperature anomalies have increased more rapidly than calorie availability, highlighting a widening gap between climate pressure and food-system adaptation.

3

Land and production volume are weak predictors on their own.

Agricultural land and production volume alone show little relationship with food availability, emphasizing that efficiency, distribution, and trade matter more than resource quantity.

4

Resilience is a balance, not a purchase.

The composite Resilience Index demonstrates that countries achieve stronger food security by balancing economic capacity, climate resilience, and self-sufficiency rather than relying on wealth alone.

ACCESS THE PROJECT

Project Resources

Live Streamlit Dashboard

https://beyond-economic-growth.streamlit.app

GitHub Repository

https://github.com/Anupriya-Segar/beyond-economic-growth

Project Notebooks

Included in the GitHub repository above

Datasets

FAOSTAT • World Bank

One Dataset. Twelve Questions. One Story.

Economic growth matters — but it isn't the whole story of food security. Climate resilience, agricultural capacity, and trade self-sufficiency matter just as much.

Thank you.